

Module-3

chemical Processes and Composting: Energy Recovery from Municipal Solid Waste. Current Issues in Solid Waste Management and Review of MSW Management Status in First List of 20 Smart Cities in the Country.

ENERGY RECOVERY FROM WASTE

Energy recovery is the process of releasing the energy stored in waste so that it can be utilised either directly or indirectly to generate heat and electricity or biofuels. Energy recovery is the process by which solid waste is converted into feedstock materials or renewable energy.

The environmental case for energy from waste versus landfill:

When considering the relative environmental benefits of landfill and energy from waste, the most important factor is their potential contribution to climate change. Different amounts of greenhouse gases would be released if the same waste was burned or buried.

The balance between the many factors that affect this is complex and much work has been done to understand it that is beyond the scope of this guide. However, there are two simple rules that can help guide our decision making on which route to follow:

- The more efficient the plant is at turning waste into usable energy the better
- The proportion of the waste that is considered renewable is key – higher renewable (biodegradable) content makes energy from waste inherently better than landfill.

Energy from waste is therefore better than landfill, providing the residual waste being used has the right renewable content and is matched with a plant that is efficient enough at turning the waste to energy. These considerations should be at the heart of any proposal.

There are rules about when energy from waste can be counted as recovery or disposal in the context of the waste hierarchy. However, if the principles above are followed then even when it is classified as disposal the environmental balance may still favour energy from waste over landfill.

Energy from waste and recycling:

There is often concern that energy from waste discourages greater recycling. Government's goal is to move waste up the hierarchy. Throughout Europe there are examples where energy from waste coexists with high recycling, ultimately delivering low landfill.

At the more local level the risk that energy from waste can compete with, not complement, recycling does exist. However, it is an avoidable risk if contracts, plants and processes are flexible enough to adapt to changes in waste arisings and composition.

Waste infrastructure has a long lifetime and care needs to be taken at the start to ensure systems can adapt to potential long term change and drive waste up the hierarchy, not constrain it. Flexibility of the overall approach to future change should therefore be another key consideration in any proposal.

Energy from waste as an energy source: Energy from waste is not just about waste management.

- The energy it produces is a valuable domestic energy source contributing to energy security.
- As a partially renewable energy source it can also contribute to our renewable energy targets which are aimed at decarbonising energy generation.
- It has the added advantage that it is non-intermittent, so it can complement other renewable energy sources such as wind or solar.

Energy outputs: Most of the energy from waste is currently produced in the form of electricity. However, more and more plants are also looking to use the heat generated. This is known as combined heat and power. More innovative technologies have the potential to also transform the waste into other energy products such as transport fuels or substitute natural gas.

The Government provides a number of different financial incentives to help drive growth in energy from waste, particularly for the more novel technologies and energy outputs beyond electricity. These along with the effective use of heat have the potential to deliver higher overall efficiency and therefore deliver the Government's goal of more energy from less waste.

Energy from Waste Infrastructure: An energy from waste solution consists of a number of components: the fuel, the plant, the location etc., and the decisions around these are interlinked.

Waste as a fuel

While waste will differ from one bag to the next, at the scale of an energy from waste plant it is possible to estimate an average composition for the waste. This will define important properties such as the energy available in the waste and the renewable content. These need to be understood to pick the best solution.

Pre-treatments

Some technologies can cope with a wide range of waste composition. However, others need more specific properties, and will therefore require pre-treatment of the waste to transform it into a refuse derived fuel. Pre-treatment requires energy which needs to be considered as part of the overall environmental assessment of the solution. The waste fuel needs to be matched to the technology in terms of both physical properties and environmental impact.

The basic process All energy from waste plants will have the same basic steps

- A reception area to receive the waste and get it ready for combustion
- A thermal treatment – this essentially releases the energy from the waste
- Conversion to a transportable form of energy – e.g. electricity, heat, fuels
- Emissions clean-up – ensuring waste gases are safe.

The overall environmental benefits will depend not only on the thermal treatment but the energy conversion technology to which it is coupled. The important factor to consider is the overall efficiency, net of any energy required to run the process.

Energy from waste technologies and outputs

The most common thermal treatment is incineration; less common are Advanced Thermal Treatments (ATT) such as gasification or pyrolysis. They each have their advantages and disadvantages, with no 'one size fits all' solution. The size and site of the plant will influence the type of technology that is appropriate.

Use of heat significantly increases the overall efficiency of the process and the environmental benefits. Sites with heat customers available should be favoured, along with technologies that can exploit this. Smaller sites may suit the ATTs which can operate at smaller scale.

The most common way to generate energy is to use hot gases from the thermal step to boil water to create steam. This is then fed into a steam turbine to generate electricity and/or used for heating. This is the only route for incineration.

Advanced thermal treatments create a mixture of products from the thermal step that still have a lot of chemical energy stored in them e.g. gases and oils. These can be burnt and used to raise steam as above. However, they also have the potential to be cleaned and burnt directly in gas engines or gas turbines, or converted to transport fuels or synthetic natural gas.

The latter routes have the potential to convert the energy from the waste more efficiently than through steam generation, which makes them attractive. However, they are technically difficult, relatively unproven at commercial scale, and some of the generated energy is used to power the process, reducing the overall benefits.

Emissions

The emissions clean-up step ensures that all the waste gases emitted from the plant meet the very tight limits placed on them by EU legislation. As a result, energy from waste plants contribute only a small fraction of both local and national particulate and other emissions.

Health impacts

The potential health implications of emissions are often a major focus of concern, hence the tight regulation of the emissions and the high priority. Government gives to the ongoing process of conducting, evaluating and disseminating high quality science. Experts have reviewed research undertaken to examine the suggested links between emissions from municipal waste incinerators and effects on health. It notes that modern, well-managed incinerators make only a small contribution to local concentrations of air pollutants.

Municipal Solid Waste (MSW) contains organic as well as inorganic matter.

The latent energy present in its organic fraction can be recovered for gainful utilisation through adoption of suitable Waste Processing and Treatment technologies. The recovery of energy from wastes also offers a few additional benefits as follows:

- (i) The total quantity of waste gets reduced by nearly 60% to over 90%, depending upon the waste composition and the adopted technology.
- (ii) Demand for land, which is already scarce in cities, for landfilling is reduced.
- (iii) The cost of transportation of waste to far-away landfill sites also gets reduced proportionately.
- (iv) Net reduction in environmental pollution.

BASIC TECHNIQUES OF ENERGY RECOVERY

- (i) Thermo-chemical conversion : This process entails thermal de-composition of organic matter to produce either heat energy or fuel oil or gas. The main technological options under this(thermo-) category include Incineration and Pyrolysis/ Gasification.
- (ii) Bio-chemical conversion: This process is based on enzymatic decomposition of organic matter by microbial action to produce methane gas or alcohol. Are preferred for wastes having high percentage of organic bio-degradable matter and high level of moisture/ water content, which aids microbial activity. The main technological options under this category is Anaerobic Digestion, also referred to as Bio-methanation.

Parameters affecting Energy Recovery:

Quantity of waste, and Physical and chemical characteristics (quality) of the waste. The actual production of energy will depend upon specific treatment process employed, the selection of which is also critically dependent upon (apart from certain other factors described below) the above two parameters.

The important physical parameters requiring consideration include:

- *Size of Constituents*: Smaller size → faster decomposition density : the high density → high proportion of biodegradable organic matter and moisture.
- *Low density* → high proportion of paper, plastics and other combustibles moisture content

- *High moisture content* → biodegradable waste fractions decompose more rapidly than in dry conditions

The important chemical parameters to be considered for determining the energy recovery potential:

- Volatile Solids
- Fixed Carbon content
- Inert,
- Calorific Value
- C/N ratio (Carbon/Nitrogen ratio)
- Toxicity

In most cases the waste may need to be suitably segregated/ processed/ mixed with suitable additives at site before actual treatment to make it more compatible with the specific treatment method.

For example, in case of Anaerobic digestion, if the C/N ratio is less, high carbon content wastes (straw, paper etc.) may be added; if it is high, high nitrogen content wastes (sewage sludge, slaughter house waste etc.) may be added, to bring the C/N ratio within the desirable range.

TECHNOLOGICAL OPTIONS

- 1 Anaerobic Digestion (AD)
- 2 Landfill Gas Recovery
- 3 Incineration
- 4 Pyrolysis/ Gasification

Anaerobic Digestion (bio-methanation)

AD process can be divided into three stages with three distinct physiological groups of micro-organisms:

Stage I: It involves the fermentative bacteria, which include anaerobic and facultative micro-organisms. Complex organic materials, carbohydrates, proteins and lipids are hydrolysed and fermented into fatty acids, alcohol, carbon dioxide, hydrogen, ammonia and sulphides.

Stage II: In this stage the acetogenic bacteria (produce acetate by anaerobic respiration) consume these primary products and produce hydrogen, carbon dioxide and acetic acid.

Stage III: It utilizes two distinct types of methanogenic bacteria. The first reduces carbon dioxide to methane, and the second decarboxylates acetic acid to methane and carbon dioxide.

Factors, which influence the Anaerobic Digestion process, are temperature, pH, nutrient concentration, loading rate, toxic compounds and mixing.

A temperature of about 35-38 degree is generally considered optimal in mesophilic zone (20-45degree) and higher gas production can be obtained under thermophilic temperature in the range of 45-60degree. Figure 1 represents the stages of Anaerobic Digestion.

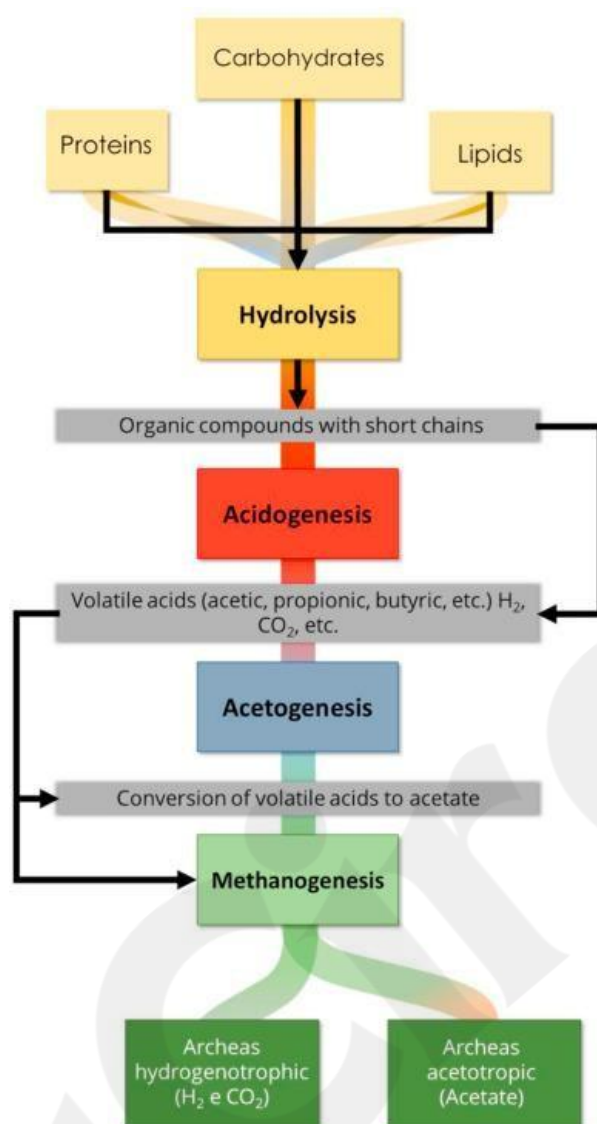


Fig 1:Anaerobic Digestion flow chart

Different Designs and Configurations of AD Systems:

- (i) *Low / Medium Solid Digestion Systems:* A large number of systems presently available worldwide for digestion of solid wastes are for low (< 10%) or medium (10-16%) solid concentrations. Some of these systems, when applied to MSW or Market Waste, require the use of water, sewage sludge or manure.
- (ii) *High Solid Continuous Digestion Systems:* Developed in late eighties principally for the organic fraction of MSW but have also been extended to other industrial, market and agricultural wastes. The digestion occurs at solid content of 16% to 40%. These systems are referred to as 'Dry Digestion' or Anaerobic Composting when the solid concentration is in the range of 25-40% and free water content is low.
- (iii) *Two Stage Digestion Systems:*
In these systems the hydrolysis, acidogenesis and acetogenesis of the waste are carried out separately from the methanogenesis stage.

(iv) *Dry Batch Digestion*: This design concept is closest to the processes occurring naturally in a landfill. The reactor containing the organic material is inoculated with previously digested waste from another reactor, sealed and allowed to digest naturally. Batchwise anaerobic process of digestion and recirculation is shown in Figure2.

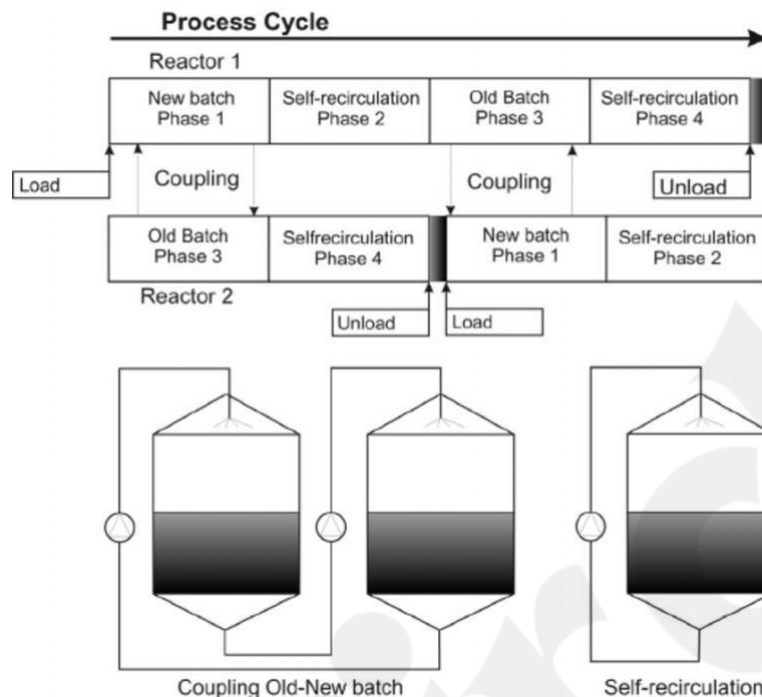


Fig 2: Batchwise Anaerobic Process

Landfill Gas Recovery

When waste is deposited in landfills, an anaerobic decomposition takes place, and landfill gas will be produced. The gas contains approximately 50% methane, which can be used for energy purposes. Extraction of gas reduces the emission of methane into the atmosphere. Figure 3 illustrates the recovery of methane from landfill and the distribution of fuel, extracted.

Using LFG helps to reduce odours and other hazards associated with LFG emissions, and it helps businesses, states, energy providers, and communities protect the environment and build a sustainable future. The main components in landfill gas recovery are: An extraction system and a utilization system.

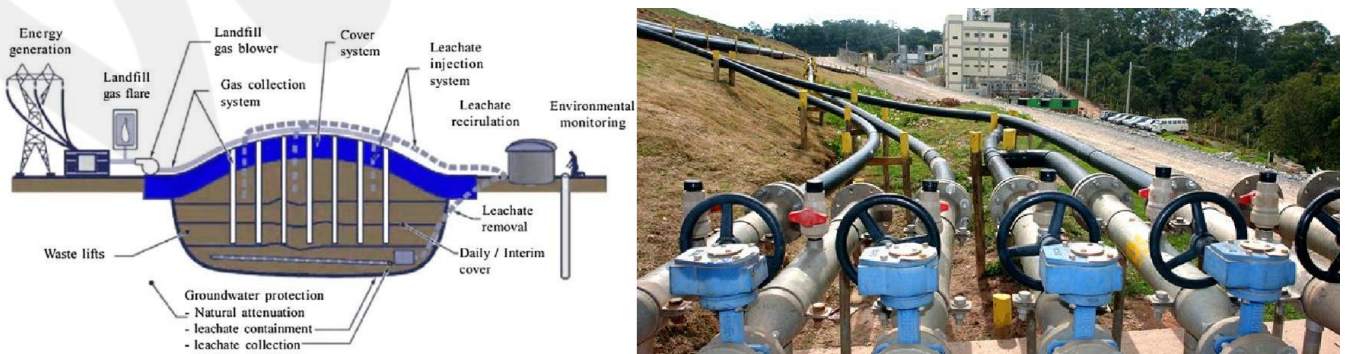


Fig 3: Landfill gas recovery process and distributed pipelines

1) Extraction System

Vertical perforated pipes, Horizontal perforated pipes, and in some cases a membrane covering for gas collection. Pump or a Compressor leading the gas into the production system. However, in a number of sites horizontal suction pipes are built in, when the waste is deposited on the landfill.

In this way the gas can easier be extracted from the very beginning of the gas production, as the gas can then be sucked out before closure/covering of the landfill. Sometimes an impermeable membrane will cover the landfill, and almost all the gas can then be collected and recovered..

2) Utilization System

The gas can be used in a gas boiler for the production of hot water for heating or process heat. Landfill gas is used as fuel in a gas engine, which drives a power generator. The following systems may be adopted for utilizing LFG.

1. Power production and Combined Heat and Power Plant (CHP Plant):

The most known use of the gas is in a gas engine running an electric generator producing power. CHP plants compared with only power production are the most efficient system for utilizing the energy from landfills.

2. Boiler System the gas is used for heating of water in a boiler system. :

The heat from some boiler systems is used in greenhouses, either by normal circulation of hot water, or by heating of air that is blown into the greenhouses. This is also a relatively simple and efficient way to use the gas.

3. Upgrading to Natural Gas Quality:

The main step in the upgrading process is the separation of methane and carbon dioxide. For this process three techniques are applied:

- Chemical Absorption
- Pressure Swing Adsorption (PSA)
- Membrane Separation

4. Use of Gas in Vehicles landfill gas is compressed and used in either compactors, refuse collection vehicles, busses or ordinary cars:

The tax system differs in the single country and is important when finding out whether the system is profitable or not. Fuel Cells may be compared to large electric batteries, which provide a means to convert the chemical bonding energy of a chemical substance directly into electricity. The difference between a battery and a fuel cell is, that in a battery, all reactants are present within the battery and are slowly being depleted during battery utilization. In a fuel cell the reactants (fuel) are continuously supplied to the cell.

Incineration

Direct burning of wastes in the presence of excess air (oxygen) at temperatures of about 800°C and above, liberating heat energy, inert gases and ash. Net energy yield depends upon:

- The density and composition of the waste;
- Relative percentage of moisture and inert materials;
- Ignition temperature;
- Size and shape of the constituents;
- Design of the combustion
- System (fixed bed/ fluidised bed), etc

The combustion temperatures of conventional incinerators fuelled only by wastes are about 760°C in the furnace, and in excess of 870°C in the secondary combustion chamber. Some modern incinerators utilise higher temperatures of up to 1650°C using supplementary fuel. The following figure 4 shows the construction of an incinerator.

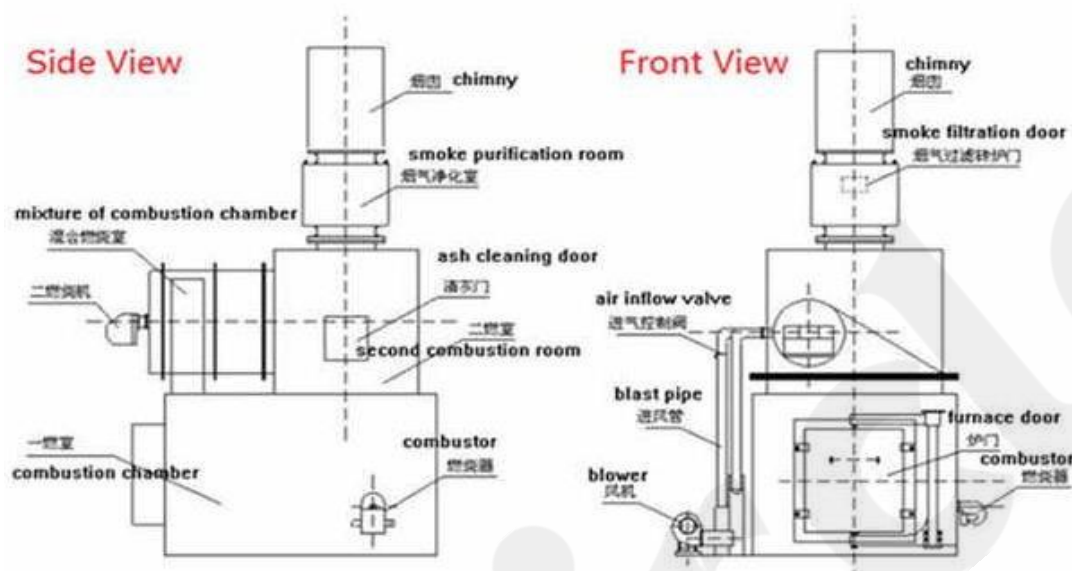


Fig 4: Incineration Process.

Basic Types of Incineration Plants

Incineration plants are categorized by combustion technology, with common types including fixed grate, moving grate, rotary kiln, and fluidized bed incinerators.

By Combustion Technology:

- **Fixed Grate and Moving Grate (Direct-Flame) Incinerators:** These are common for large waste treatment plants, burning waste in a single combustion chamber with excess air.
 - **Fixed Grate:** Waste is burned on a stationary grate.
 - **Moving Grate:** Waste is moved along a series of moving grates.
- **Rotary Kiln Incinerators:** These use a rotating kiln for high-temperature combustion, suitable for various waste types.
- **Fluidized Bed Incinerators:** Waste is suspended in a bed of hot, moving particles, promoting efficient combustion.
- **Pyrolytic Chamber Incinerators:** These use a pyrolytic chamber and a post-combustion chamber for more complete combustion.
- **Liquid Injection Incinerators:** These are used for incinerating liquid wastes.
- **Multiple Hearth Incinerators:** These use multiple hearths for waste processing.
- **Catalytic Combustion Chamber:** These use a catalyst to reduce emissions.
- **Waste Gas Flare Incinerator:** These incinerate waste gases.

Other Classifications:

- **Mass-burn systems:** Incinerate all waste, typically large facilities.
- **Refuse-derived fuel (RDF) systems:** Process waste to create a fuel source for incineration.

- **Hazardous Waste Incinerators:** Designed for burning hazardous materials at high temperatures to destroy contaminants.
- **Medical Waste Incinerators:** Specially designed for burning medical waste, including infectious and non-infectious materials.

Pyrolysis/ Gasification

Pyrolysis is also referred to as destructive distillation or carbonization. It is the process of thermal decomposition of organic matter at high temperature (about 900°C) in an inert atmosphere or vacuum, producing a mixture of combustible Carbon Monoxide, Methane, Hydrogen, Ethane [CO , CH_4 , H_2 , C_2H_6] and non-combustible Carbon Dioxide, water, Nitrogen [CO_2 , H_2O , N_2] gases, pyrolygneous liquid, chemicals and charcoal. The process of pyrolysis is shown in figure 5.

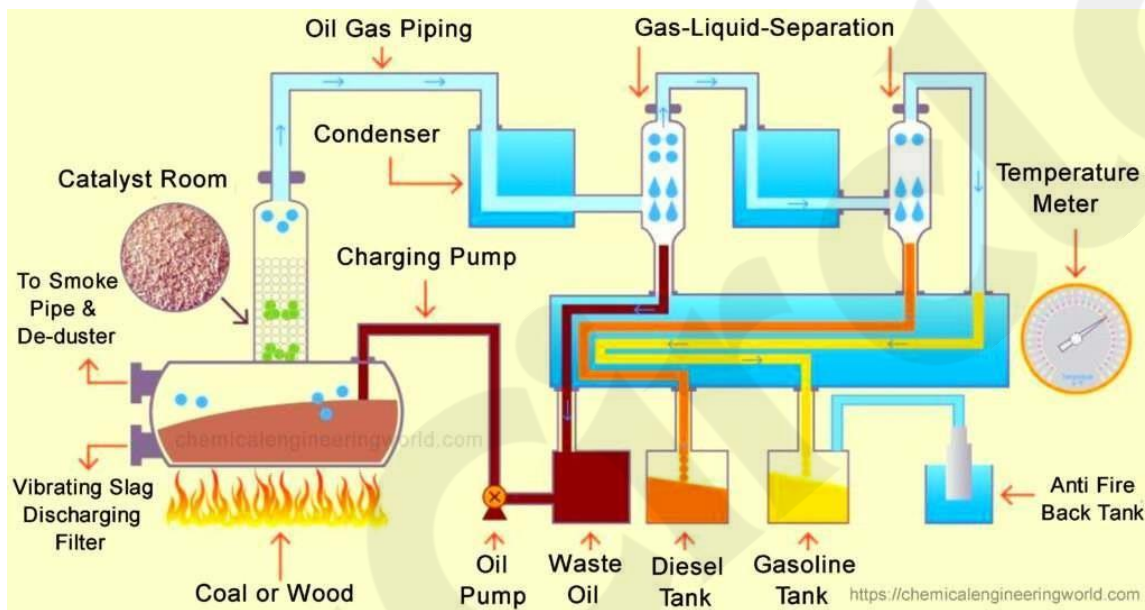


Fig 5: Pyrolysis process

Gasification involves thermal decomposition of organic matter at high temperatures as shown in figure 6 in presence of limited amounts of air/ oxygen, producing mainly a mixture of combustible and non-combustible gas (carbon Monoxide, Hydrogen and Carbon Dioxide).

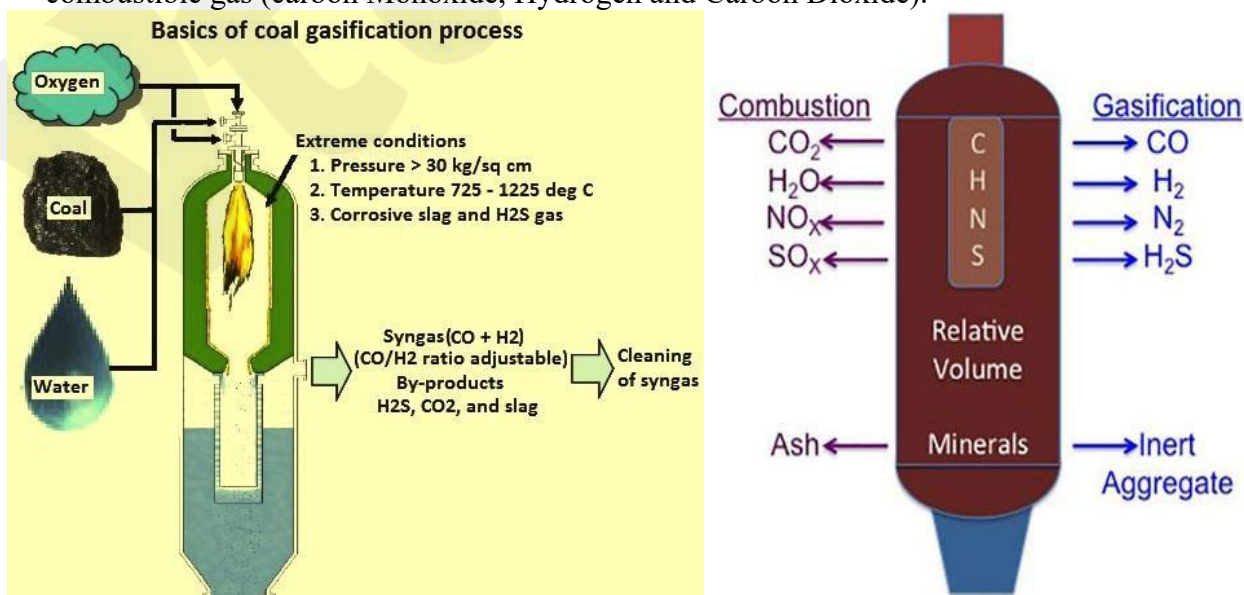


Fig 6: Gasification reactions and the by-products.

This process is similar to Pyrolysis, involving some secondary/ different high temperature ($>1000^{\circ}\text{C}$) chemistry which improves the heating value of gaseous output and increases the gaseous yield (mainly combustible gases $\text{CO}+\text{H}_2$) and lesser quantity of other residues.

Current Issues in Solid Waste Management

Solid waste management is a critical challenge worldwide, and several pressing issues affect its efficiency and sustainability. Here are some of the current issues in solid waste management:

- 1. Increasing Waste Generation:** Rapid urbanization and population growth have led to an exponential rise in solid waste generation. Many cities struggle to keep up with collection, transportation, and disposal.
- 2. Inadequate Waste Segregation:** Lack of proper waste segregation at the source leads to inefficient recycling and landfill overuse. Public awareness and enforcement of segregation policies remain a challenge.
- 3. Overflowing Landfills & Limited Space:** Many existing landfills are nearing capacity, causing environmental and health hazards.
Finding new landfill sites is difficult due to space constraints and public opposition (NIMBY – "Not In My Backyard" syndrome).
- 4. Inefficient Recycling & Low Public Participation:** Many recyclable materials end up in landfills due to lack of awareness, infrastructure, or incentives.
Informal waste pickers play a key role but often lack support and recognition.
- 5. Hazardous & Electronic Waste Mismanagement:** E-waste and hazardous materials like batteries, chemicals, and medical waste require specialized disposal.
Many developing countries lack proper regulations and disposal facilities for such waste.
- 6. Marine & Plastic Pollution:** Plastic waste, especially single-use plastics, continues to pollute oceans, rivers, and cities.
Microplastics have become a major concern for marine life and human health.
- 7. Inefficient Waste-to-Energy Technologies:** While waste-to-energy plants can reduce landfill dependency, many countries struggle with high costs and technology adoption.
Emissions from incineration plants raise environmental concerns.
- 8. Policy & Regulatory Challenges:** Many regions lack strict enforcement of waste management policies. Corruption and inefficient governance hinder proper waste management practices.
- 9. Climate Change Impact on Waste Management:** Extreme weather events like floods can disrupt waste collection and landfill operations. Decomposing waste in landfills releases methane, a potent greenhouse gas, contributing to global warming.
- 10. COVID-19 Waste Management Issues:** The pandemic led to an increase in medical waste, including masks, gloves, and PPE kits. Many municipalities struggled to handle the sudden surge in biomedical waste safely.

Possible Solutions:

- ✓ Promoting waste segregation at the source.
- ✓ Encouraging circular economy and zero-waste policies.

- ✓ Expanding recycling infrastructure and waste-to-energy initiatives.
- ✓ Strengthening policies and enforcing strict waste management laws.
- ✓ Raising public awareness through education and incentives for responsible waste disposal.

REVIEW OF MSW MANAGEMENT STATUS IN FIRST LIST OF 20 SMART CITIES IN THE COUNTRY

A review of MSW (Municipal Solid Waste) management in the first list of 20 Indian smart cities reveals a mixed picture, with some cities showing progress in collection and segregation, but significant gaps remain in proper treatment and disposal, highlighting the need for improved infrastructure, public awareness campaigns, and effective waste management strategies to achieve true "smart city" status across all aspects of waste management. Some of key findings are:

- **Uneven progress:**

While some cities like Surat, Indore, and Visakhapatnam have made strides in waste collection and segregation initiatives, others still struggle with basic collection systems and lack proper treatment facilities.

- **Focus on collection:**

Most cities prioritize waste collection, but the treatment and recycling aspects, including composting and energy recovery, are often underdeveloped.

- **Challenges with segregation:**

Inadequate source segregation at household level remains a major challenge, impacting the efficiency of waste processing.

- **Informal sector involvement:**

Many cities rely heavily on informal waste pickers for collection, requiring better integration and support to ensure proper waste management.

- **Landfill dependence:**

Despite efforts to reduce reliance on landfills, many cities still primarily dispose of waste in open dumps, leading to environmental concerns.

Positive examples of smart cities in waste management:

- **Surat:** Recognized for its focus on waste segregation and efficient collection systems.
- **Indore:** Demonstrates good waste management practices with initiatives like door-to-door collection and composting.
- **Visakhapatnam:** Implementing advanced waste treatment technologies like RDF (Refuse Derived Fuel) generation.
- **Areas needing improvement:**
 - **Bhubaneswar:** Requires better waste collection infrastructure and improved recycling facilities.
 - **Jaipur:** Needs more awareness campaigns to promote source segregation.
 - **Ahmedabad:** Challenges in managing large volumes of waste and improving landfill management.

The key areas for improvement for better waste management in smart cities are:

- **Infrastructure development:**

Investing in modern waste collection vehicles, transfer stations, and treatment plants like composting units and waste-to-energy facilities.

- **Public awareness campaigns:**

Educating citizens about proper waste segregation and responsible disposal practices.

- **Policy and regulatory framework:**

Strengthening existing regulations and enforcement to ensure compliance with waste management guidelines.

- **Collaboration with the informal sector:**

Integrating informal waste pickers into the formal waste management system.

Overall, while some cities in the first smart city list are taking positive steps towards improved MSW management, a comprehensive approach is needed to address the existing gaps and achieve sustainable waste management practices across all selected cities.

Integrated Waste Management in smart cities:

In India's 20 smart cities, waste management initiatives primarily focus on utilizing technology like IoT sensors in smart bins to monitor fill levels, enabling efficient collection routes, promoting source segregation through color-coded bins, and implementing door-to-door collection systems with a strong emphasis on awareness campaigns to encourage the "Reduce, Reuse, Recycle" approach; notable cities include Indore, Visakhapatnam, Bhopal, and Udaipur, which are leading the way with pilot projects in waste segregation and advanced collection methods. The aspects of waste management in Indian smart cities:

- **Smart bins with sensors:**

Bins equipped with IoT sensors to monitor fill levels, allowing optimized collection schedules and reducing unnecessary truck trips.

- **Source segregation:**

Promoting citizen participation by providing separate bins for dry and wet waste, encouraging proper segregation at the source.

- **Door-to-door collection:**

Implementing efficient door-to-door collection systems to ensure convenient waste disposal.

- **Waste-to-energy plants:**

Utilizing waste to generate energy through incineration plants in some cities.

- **Recycling facilities:**

Setting up recycling centres to process recyclable materials like plastic, paper, and glass.

- **GIS mapping:**

Utilizing Geographic Information Systems to map waste generation hotspots and optimize collection routes.

- **Community awareness campaigns:**

Educating citizens about the importance of waste segregation and responsible waste management practices.

Some prominent smart cities leading in waste management:

- **Indore:** Considered one of the cleanest cities in India, with a robust waste segregation system and awareness campaigns.
- **Visakhapatnam:** Implementing pilot projects with door-to-door collection and color-coded bins.
- **Bhopal:** Initiating projects to improve waste collection and treatment facilities.
- **Udaipur:** Focusing on source segregation and awareness programs

Challenges in smart city waste management:

- **Lack of citizen participation:** Insufficient awareness and engagement in proper waste segregation.

- **Inadequate infrastructure:** Insufficient waste processing facilities and collection vehicles in some cities.
- **Enforcement issues:** Difficulty in enforcing waste management regulations and penalties for non-compliance.

Pathways to sustainable waste management in Indian Smart Cities

Administrative structure

The administrative structure for governing the SWM activities includes three levels -centre, state, and local. The Ministry of Environment, Forest, and Climate change (MoEF & CC) is the authority for reviewing, monitoring, and implementing SWM rules at the national level. The MoEF & CC constitutes a monitoring committee for evaluating the status of solid waste management yearly. The CPCB is the coordinating authority between the central and state government levels. CPCB reviews, monitors, and publishes environmental standards and approves technologies. The State Pollution Control Board (SPCB) is the coordinating body between the CPCB and the council authorities. They have the same duties as the CPCB at the state level. They also regulate the interstate transfer of waste. At the council level (urban local body), the state government appoints an Indian Administrative Service (IAS) rank officer designated as Commissioner. The sanitation section under the public health division is the authority for waste management activities. Revenue, finance, education, engineering, and town planning divisions also extend services to develop SWM systems. The town planning section is instrumental in the planning, identifying, and allocating land for the setting up of solid waste processing, treatment, and disposal facilities. The civil engineering section will look after construction activities; the mechanical division will look after transport activities and heavy vehicles involved in SWM activities. The finance and revenue wings are responsible for budget allocation and auditing, collection of taxes, and levies. The education division is engaged in conducting awareness programs on source segregation, treatment of waste, and health effects due to improper waste management. The figure 1 shows the organization or administrative structure of the Municipal Corporation.

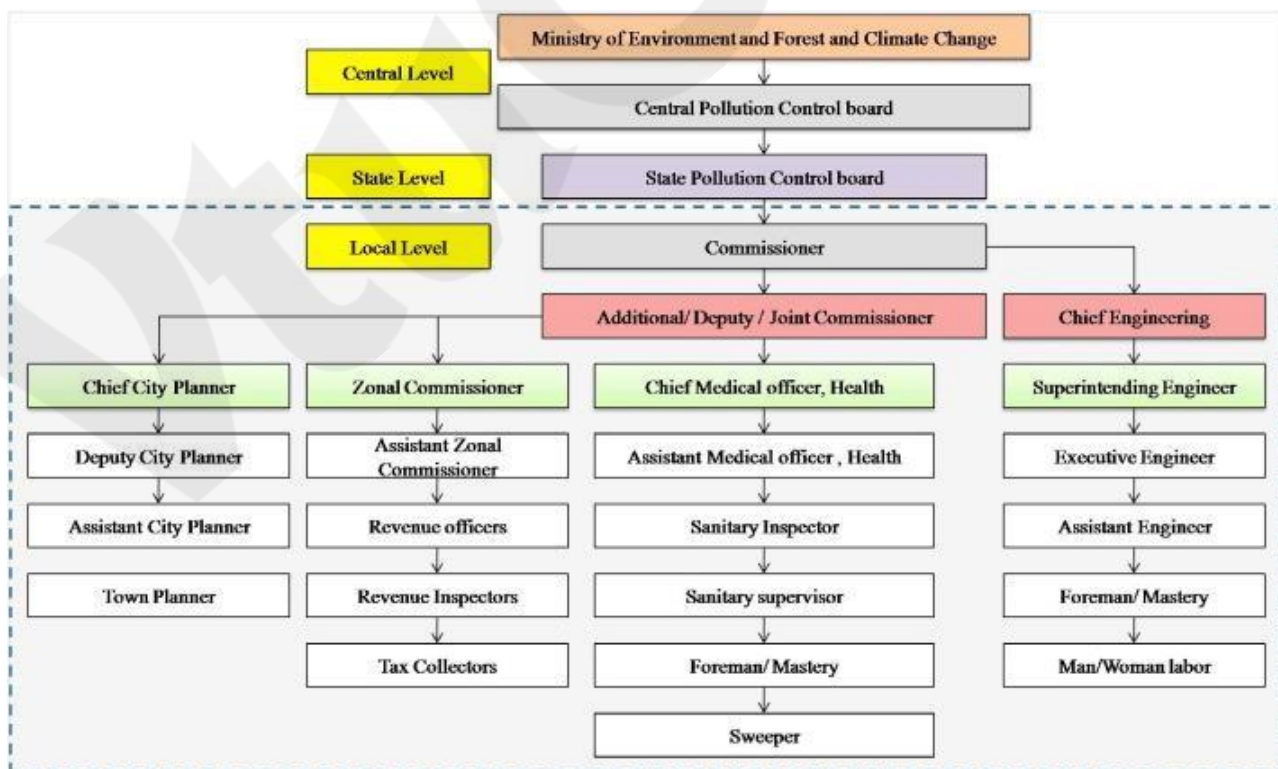


Figure 1: Organization or administrative structure of the Municipal Corporation.

Indicative pathways for designing a sustainable waste system

Studies performed in the selected Indian smart cities revealed that effective integration of the new solid waste management rules with the government objectives and diagnostic analysis of existing systems provides insights into futuristic strategies. The study's indicative strategies provide insight into developing an integrated solid waste management plan for different urban cities across the globe. Source segregation of the waste is the key to the development of good waste treatment and processing system. Scientific approach integrated with the technologies and effective administration is the key for transforming the existing waste management systems to develop smart and liveable cities. Integration of the public and private sector by creating special purpose vehicles will provide scope for investments. Maintenance of data records will provide scope for developing a waste management index for systematic monitoring of the implementation of provisions developed by the regulatory norms. The figure 2 represents the solid waste management systems in the study areas.

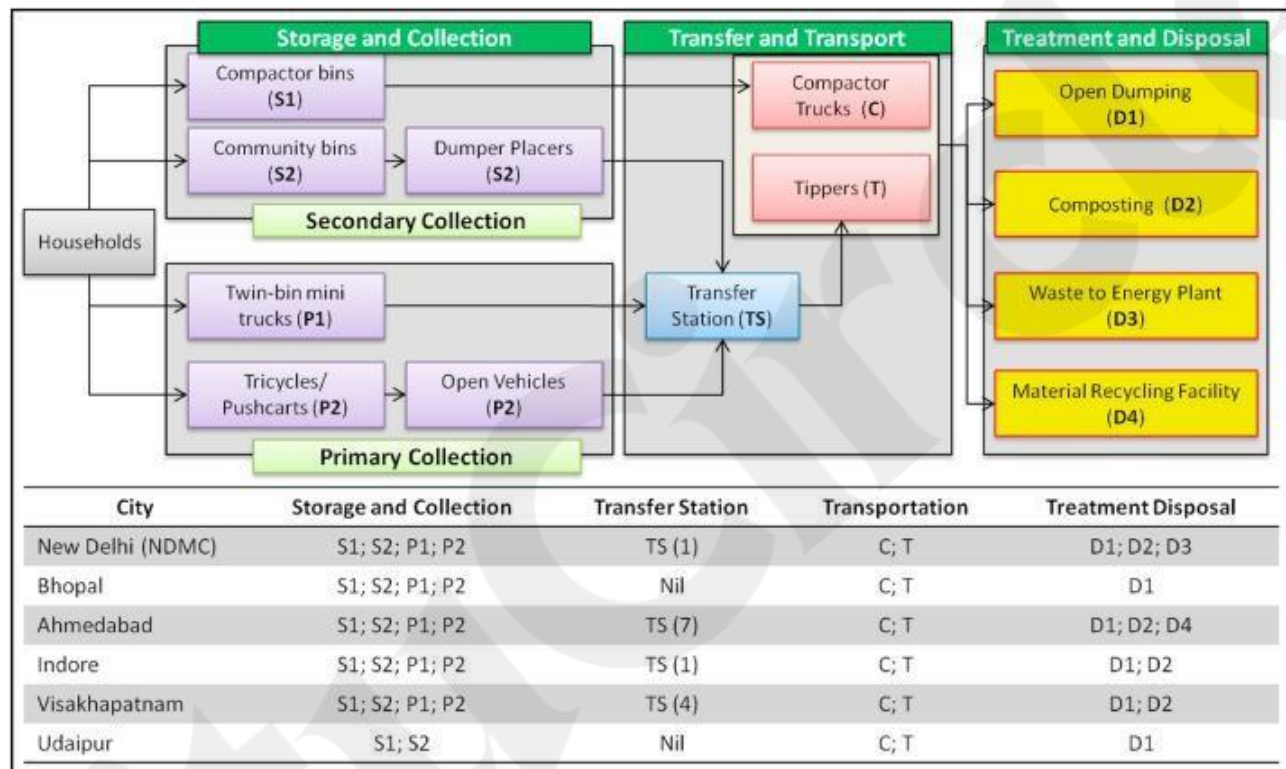


Fig 2: The solid waste management systems in the study areas.

Financial provisions for waste management systems

The economic component is an essential parameter for the decision-making process to a ULB in a developing country like India. Councils allocate 70–80% of the budget for waste collection and transportation systems. Inefficient source segregation and storage mechanisms pose a critical challenge to the local authorities. Development of infrastructure is required to facilitate the constant increase in the quantities of waste. In the Indian scenario, mixed solid waste is reaching the waste treatment units. Mixed waste reduces the quality of the output product leading to a decrease in profit, resulting in an infeasible economic situation for the treatment plants. Collaboration under public-private partnerships and corporate social responsibility also contribute to the development of waste management systems.

The circular economy is a new way of thinking for sustainable growth in developing countries. The basic concept behind the linear economy is: “take, make, consume, and discard. In the linear economy, products are used until they are discarded and disposed of as waste to the environment, and disposed waste is not used further. In the circular economy, waste management practices are based on the 3R approach “Reduce,

Reuse, and Recycle,” where waste products are recycled and reused rather than disposed of to the environment. In the circular economy, waste from one process becomes a valuable input for other operations. Waste management through circular economy approaches could be a boon in terms of revenue generation and employability opportunities for developing countries. The circular economy concept requires improved design and production practices to use waste material as a resource efficiently.

Application of the systems engineering approaches

In a smart city, the waste management systems should be planned, designed, optimized, and implemented to ensure sustainable waste management systems. Collaborative planning, infrastructure development, capacity building, database management, and implementation play a vital role in developing an integrated waste management system. A system engineering framework combining decision support systems (Multi-Criteria Decision Modeling and Knowledge-Based systems, Remote Sensing and Geographical Information Systems, Life Cycle Assessment, Information and Communication Technologies (ICT), and Internet of Things (IoT) applications provide a multidisciplinary approach for the urban planners

Development of solid waste management performance index

The development of SWM systems requires a strategic and sustainable approach that emphasizes environmental performance, financial viability, and social equity. A system is a time-targeted approach involving multiple stakeholders during different stages of execution. Assessment of the operations at regular intervals, both qualitative and quantitative, provides an overview of the project based on the site-specific conditions. These assessment reports give a holistic overview, equipping the regulatory and local authorities during decision-making. Swachh Sarvekshana is a service level benchmarking (SLB) exercise conducted by the Government of India annually since 2016 to assess water and sanitation systems. Solid waste management is one of the significant parameters in this SLB. The indicators for SWM include household level coverage of SWM services, the efficiency of collecting MSW, the extent of segregation and recovery, the scientific disposal of municipal solid waste, efficiency in customer complaints' redressal, collection SWM charges, and cost recovery in SWM services. A detailed report will be developed initially and submitted to the Ministry of Housing and Urban Affairs in this process. A team of experts will assess the documented proofs and review the systems physically based on the report provided by the ULB. The performance index calculation includes a review from experts, field/site visits, and citizen feedback.

Implementation of this mechanism provides a peripheral overview of the systems in the ULB. Hence to have precise quality control and assurance on the MSW practices, a Solid Waste Performance Index (SWPI) is required. SWPI will include environmental, economic, and social components for each unit process involved in SWM. Development of datasheets for each unit operation of SWM, including all three elements, provides scope for identifying indicators. Goal and targets are designed based on the identified indicators specific to the ULB. The data sheets are made available to the ULBs, service providers, and stakeholders involved in the SWM. A detailed report is submitted based on the specified time intervals. Based on the collected data, the scores will be developed for each indicator. Final performance index The weighting process will be used to arrive at the final performance index.

CASE STUDIES:

Solid Waste Management (SWM) by Smart Cities, with a particular focus on Bhubaneswar and Rourkela in Odisha, India. The report critiques the implementation of SWM initiatives within the Smart City Mission guidelines.

1. Objectives of Smart Cities & Institutional Mechanisms

The Smart City Mission aimed to provide core infrastructure and a sustainable environment. The document states: "The objective of Smart City Mission was to promote cities that provide core infrastructure and give a decent quality life to its citizens, a clean and sustainable development." To implement the mission, Special Purpose Vehicles (SPVs) were envisioned at the city level to plan, appraise, approve, and manage smart city projects. These SPVs are headed by Chief Executive Officers and nominees of central and state governments.

3. Strategic & Financial Action Plans

As per the guidelines, Smart City Proposals (SCPs) should have consisted of strategic action plans for area developments. The SCP should include a financial plan identifying internal and external funding sources over an 8-10 year period.

4. Citizen-Driven Action Plan SCPs were also intended to have a citizen-driven action plan to engage citizen participation in waste management in the smart cities.

5. Implementation of SWM Component & Waste Generation

The implementation of the SWM component of the Smart City Mission is the responsibility of the SPV and requires a dedicated and sustainable revenue stream. The document presents data on waste generation in Bhubaneswar Smart City Ltd (BSCL) and Rourkela Smart City Ltd (RSCL) from 2011-12 to 2014-15 and from 2015-16 to 2019-20, focusing on solid waste and C&D waste. Data indicates substantial amounts of waste disposed of in landfills without processing. The report highlights the "hierarchy of 3Rs... Reduce-to reduce unnecessary waste generation, Reuse-to use again and Recycle- convert unwanted things into and marketable recycled products."

6. Diversion of Funds and Non-Collection of User Charges

Smart City Mission guidelines recommended waste management in cities like waste to energy and fuel, waste to compost, and water to be treated and recycling and reduction of Construction and demolition waste. The report notes instances of funds released for SWM projects ("Waste-Lets-Recycle") being diverted or used ineffectively. For example, BSCL released funds to BMC in two phases toward hiring charges of excavator, water tanker, vehicles, mini truck, transportation of solid waste, supply of bleaching powder, purchase of dustbin, conservancy materials etc., without accomplishing the objective of "Waste-Lets-Recycle".

Moreover, the audit reveals that authorized user charges for SWM were not consistently collected by Bhubaneswar and Rourkela Municipal Corporations. "Audit observed that no such user charges were collected by Bhubaneswar and Rourkela Smart cities."

7. "Swachha Sarvekshan" Ranking

The "Swachha Sarvekshan" is a ranking assessment of cities based on sanitation, garbage-free status, hygiene, etc. Bhubaneswar was recognized as a leading city in using technology for resolving urban challenges. However, the report indicates discrepancies, stating, "It has been observed that the

veracity of the survey and rankings by MoHUA was incomprehensible as the situation of SWM in these smart cities was not satisfactory as evident from the following."

8. Monitoring Deficiencies

- **Lack of State Level Monitoring:** A high-powered steering committee (HPSC) at the State level, chaired by the Chief Secretary, is meant to oversee the mission. However, "However HPSC had not taken any review meetings for the projects executed by these smart cities. It indicates poor support to effective implementation of Smart City projects."
- **Deficient Monitoring at City Level:** Similarly, the city-level advisory forum, advising the SPV, lacked proper monitoring of SWM activities, and community participation in smart city development was missing. "Audit observed that no such city level forum committee was established in both the smart cities to monitor waste management as there was no mayor in these corporations."

9. Key Concerns & Conclusions

The report raises concerns about the following:

- Lack of strategic and financial planning for SWM projects.
- Insufficient citizen participation in waste management initiatives.
- Failure to achieve targets for waste processing and landfill diversion.
- Diversion and misuse of funds allocated for SWM.
- Inconsistent collection of user charges.
- Inadequate monitoring at both state and city levels.
- Discrepancies between "Swachha Sarvekshan" rankings and the actual state of SWM.

The findings suggest that while the Smart City Mission aimed to improve SWM, its implementation in Bhubaneswar and Rourkela faced significant challenges related to planning, execution, financial management, community engagement, and oversight.

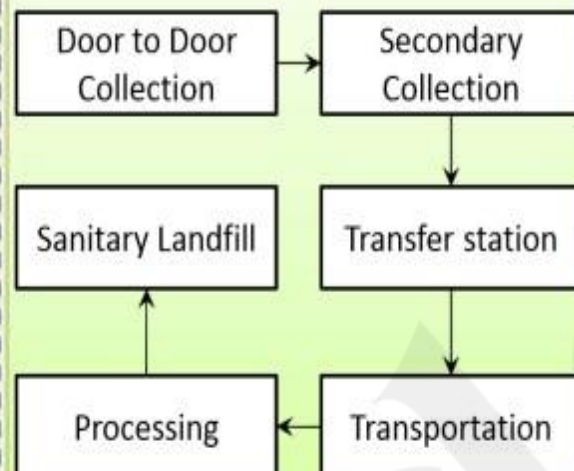
Scenarios in few cities for waste management

Action Plans for implementation of SWM 2016 rules

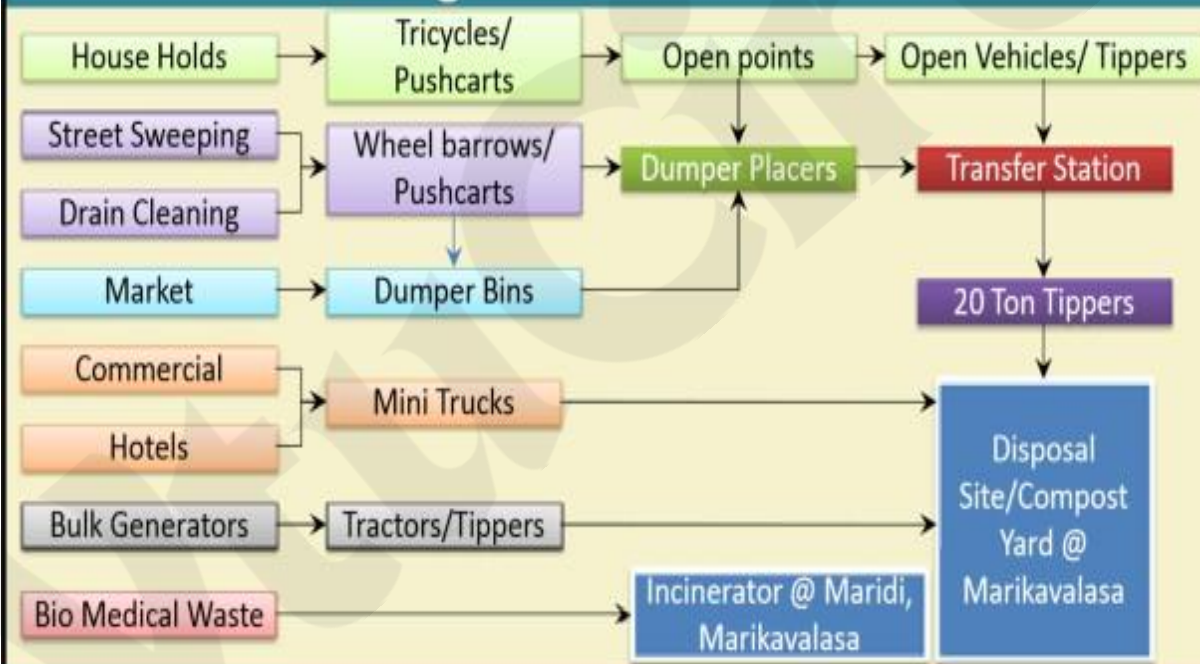
- Roadmap for Zero Waste Ahmedabad by 2031
- Swachhata Jan Model (SJM): Waste segregation at source and inclusion of waste pickers in main stream
- Setting up of Solid Waste Recycling Centre (Material Recovery Facility)
- Scientific Closure of existing Pirana Dump Site
- Implementation of Waste to Energy techniques for treatment of waste

Greater Visakhapatnam Municipal Corporation

Feature	Description
Name of the City	Visakhapatnam, Andhra Pradesh
Area	534 sq Km
No. of zones	6
No. of Wards	72
Solid waste (TPD)	1200
Population	63 lakhs (as on December 2016)



Waste Management Flow chart of GVMC

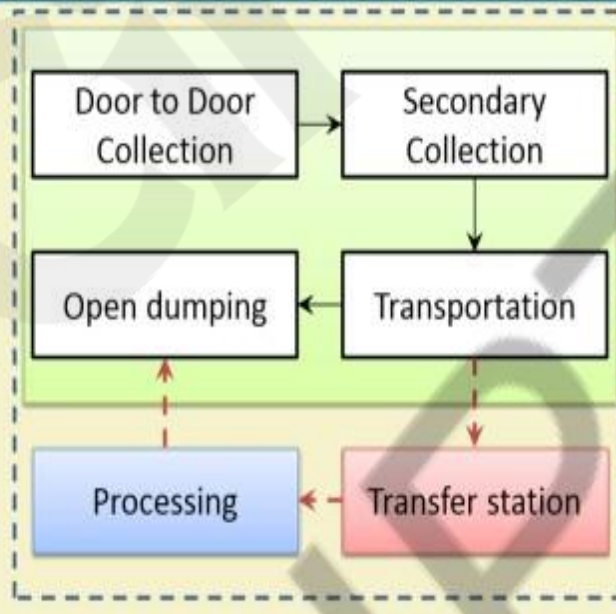


Action Plans for implementation of SWM 2016 rules

- Source Segregation of waste has been introduced in 6 wards as Pilot Project
- Emphasis is being laid on mechanization of garbage handling
- Door to door waste collection has been started in 2 wards from each zone.
- Regular Sweeping (Night sweeping of main roads) and Drain Cleaning
- New landfill sites are being developed as scientific landfill sites.
- Development of Waste to Energy plant in collaboration with Jindal Group for production of energy
- Bio-mining of waste at dump yard in collaboration with Coramandel Groups

Indore Municipal Corporation

Feature	Description
Area	130 Sq. Kms
Population (inLacks)	19,60,630
Solid Waste Generated	1000 Tons
Percentage of Waste Collected	70 - 75 %
Amount of Waste processed	25 - 30 %



Indore Municipal Corporation



Central Workshop for Operation and Maintenance of the vehicles

Training and Learning Center developed by the Municipal Corporation for creating awareness among the citizens



Action Plans for implementation of SWM 2016 rules

- The IMC has taken a step towards door to door collection.
- As a part of the program the Solid waste team is creating awareness among the citizens.
- Waste collected from city is disposed at Devguradia trenching ground existing landfill site that is 18 km away from Indore City.
- IMC in collaboration with Private organizations is planning to start up source segregation for residential groups and welfare associations.

Bhopal Municipal Corporation

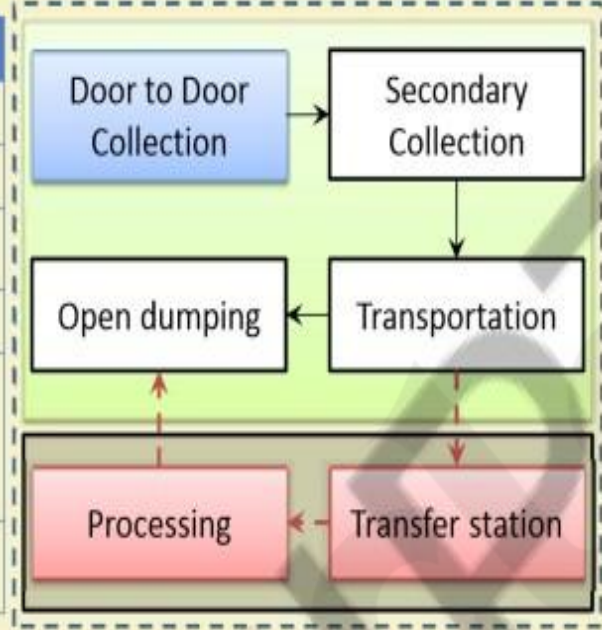
Particulars	Description		
Name of the city	Bhopal, Madhya Pradesh	Door to Door Collection	Secondary Collection
Area (sq. km)	258		
No. of Zones	14	Open dumping	Transportation
No. of Wards	66		
Solid waste (TPD)	550	Processing	Transfer station
Population (Lacs)	14		

Action Plans for implementation of SWM 2016 rules

- Cluster based solid waste system is developed for collection and transfer of waste
- Waste processing plant has to be commission and run by M.P. Agro state organization to develop Bio fertilizer of capacity 100MT/day.
- Bio-methanation plant is under construction at the market yard.
- Sanitary landfill is being developed at Agnampudi for disposal of waste.

Udaipur Municipal Corporation

Particulars	Description
Name of the city	Udaipur, Rajasthan
Area (sq. km)	64
No. of Zones	5
No. of Wards	55
Solid waste (TPD)	230 (Excluding Industrial & Hospital waste)
Population (Lacs)	4.68

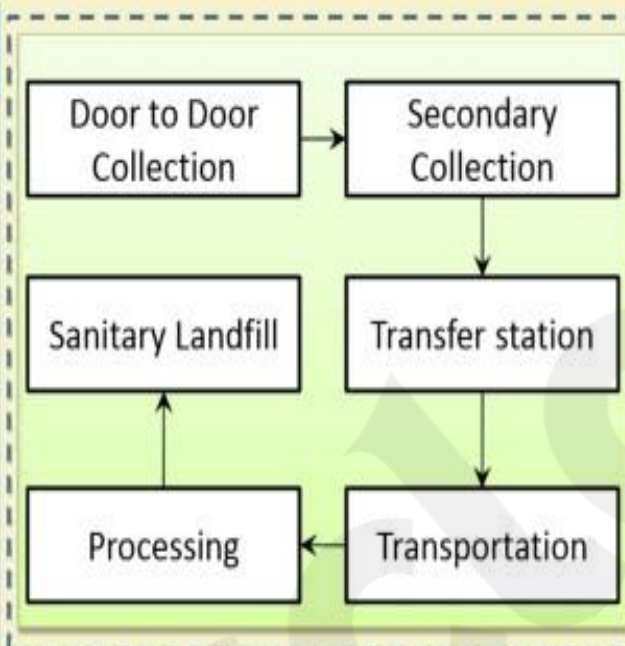


Action Plans for implementation of SWM 2016 rules

- The UMC has initiated steps towards door to door collection and Source Segregation of waste as Pilot Project
- Regular Sweeping and Drain Cleaning as the city is a tourist place
- The UMC in association with Hindustan Zinc Ltd., is planning to start up composting and Waste to Energy plants.
- Emphasis is being laid on mechanization of garbage handling

New Delhi Municipal Corporation (NDMC)

Particulars	Description
Name of the city	NDMC, Delhi
Area (sq. km)	42.8
Solid waste (TPD)	250
Population (lacs)	4.5
Landfill (TPD)	170
Composting (TPD)	80



Action Plans for implementation of SWM 2016 rules

- **Collection of garbage:**
 - Door to door collection of garbage has been started at 51 places. NDMC envisages expanding this scheme in its entire area. 1100 pairs of litterbins have been provided at busy places.
- **Collection and transportation of municipal solid waste:**
 - 1600 twin bin trolleys (Blue and Green) along with garbage stations having segregation facilities will be provided.
 - The garbage will be transported in mechanized covered vehicles to the disposal site. The garbage shall be recycled at the disposal site by installing modern compost plant and recycling plant.
- **Disposal of horticulture waste:**
 - Efforts are being made to compost the horticulture waste in situ by local composting in the NDMC nurseries.

List of First 20 Smart Cities of India (Upcoming)

The following are names of urban areas which will be developed as the top 20 smart cities of India.

1. Bhubaneswar, Odisha

Indicator	City (Municipal Corporation)
Total Population	840834
Total Area	135 km ²
Literacy Rate (%)	91.89
Per Capita Income (Rs.) at 2004-05 constant price	33312
Unemployment Rate, 2011-12	4.27
% of households with access to tap water	62.96
% of households with access to electricity	86.53
% of households having toilet facilities within premises	76.47
% of household Waste water outlet connected to drainage	68.95
Type of sewerage system	Underground sewerage system
Type of solid Waste system	Door to Door
% of households with access to computer/laptop with internet	16.05
% of households with access to computer/laptop without internet	14.80
% of households with access to mobile phones	64.74
No. of Hospitals per 1,00,000 people	1
No of Schools per 1,00,000 people	57
No of Colleges per 1,00,000 people	9
Swatch Bharat ranking (2015)	331

2. Pune, Maharashtra

Indicator	City (Municipal Corporation)
Total Population	3124458
Total Population of UA (if)	5057709
Area (sq. km)	276.4 km ²
Literacy Rate (%)	89.56
Per Capita Income (Rs.) at 2004-05 constant price	88341
Unemployment Rate, 2011-12	3.29
% of households with access to tap water	99.20
% of households with access to electricity	98.31
% of households having toilet facilities within premises	89.45
% of household Waste water outlet connected to drainage	98.04
Type of sewerage system	Underground sewerage system
Type of solid Waste system	Door to Door
% of households with access to computer/laptop with internet	22.01
% of households with access to computer/laptop without internet	16.60
% of households with access to mobile phones	64.36
No. of Hospitals per 1,00,000 people	0.03
No of Schools per 1,00,000 people	90
No of Colleges per 1,00,000 people	9
Swatch Bharat ranking (2015)	31

3. Jaipur, Rajasthan

Indicator	City (Municipal Corporation)
Total Population	3046163

Area	484.64 sq. km.
Literacy Rate (%)	83.33
Per Capita Income (Rs.) at 2004-05 constant price	3046163
Unemployment Rate, 2011-12	4.09
% of households with access to tap water	88.85
% of households with access to electricity	97.82
% of households having toilet facilities within premises	89.45
% of household Waste water outlet connected to drainage	90.94
Type of sewerage system	Underground sewerage system
Type of solid Waste system	Door to Door
% of households with access to computer/laptop with internet	12.42
% of households with access to computer/laptop without internet	13.84
% of households with access to mobile phones	66.12
No. of Hospitals per 1,00,000 people	1
No of Schools per 1,00,000 people	67
No of Colleges per 1,00,000 people	12
Swatch Bharat ranking (2015)	370

4. Surat, Gujarat

Indicator	City (Municipal Corporation)
Total Population	4467797
Total Population of UA (if)	4591246
Area	335.82
Literacy Rate (%)	87.89
Per Capita Income (Rs.) at 2004-05 constant price	NA

Unemployment Rate, 2011-12	0.29
% of households with access to tap water	80.50
% of households with access to electricity	98.88
% of households having toilet facilities within premises	93.55
% of household Waste water outlet connected to drainage	96.26
Type of sewerage system	Underground sewerage system
Type of solid Waste system	Door to Door
% of households with access to computer/laptop with internet	5.07
% of households with access to computer/laptop without internet	8.12
% of households with access to mobile phones	71.42
No. of Hospitals per 1,00,000 people	0.02
No of Schools per 1,00,000 people	29
No of Colleges per 1,00,000 people	2
Swatch Bharat ranking (2015)	63

5. Kochi, Kerala

Indicator	City (Municipal Corporation)
Total Population	602046
Total Population of UA (if)	2119724
Area	107.13 sq. km.
Literacy Rate (%)	97.36
Per Capita Income (Rs.) at 2004-05 constant price	63599
Unemployment Rate, 2011-12	4.45
% of households with access to tap water	95.55
% of households with access to electricity	99.08

% of households having toilet facilities within premises	94.62
% of household Waste water outlet connected to drainage	87.41
Type of sewerage system	Underground sewerage system
Type of solid Waste system	Door to Door
% of households with access to computer/laptop with internet	20.87
% of households with access to computer/laptop without internet	14.86
% of households with access to mobile phones	54.12
No. of Hospitals per 1,00,000 people	NA
No of Schools per 1,00,000 people	NA
No of Colleges per 1,00,000 people	NA
Swatch Bharat ranking (2015)	409

6. Ahmedabad, Gujarat

Indicator	City (Municipal Corporation)
Total Population	5577940
Total Population of UA (if)	6357693
Area	468.92 sq. km.
Literacy Rate (%)	88.29
Per Capita Income (Rs.) at 2004-05 constant price	NA
Unemployment Rate, 2011-12	0.86
% of households with access to tap water	80.12
% of households with access to electricity	98.56
% of households having toilet facilities within premises	92.42
% of household Waste water outlet connected to drainage	93.10

Type of sewerage system	Underground sewerage system
Type of solid Waste system	Door to Door
% of households with access to computer/laptop with internet	10.32
% of households with access to computer/laptop without internet	11.41
% of households with access to mobile phones	62.51
No. of Hospitals per 1,00,000 people	0.09
No of Schools per 1,00,000 people	33
No of Colleges per 1,00,000 people	2
Swatch Bharat ranking (2015)	79

7. Jabalpur, Madhya Pradesh

Indicator	City (Municipal Corporation)
Total Population	1055525
Total Population of UA	1268848
Area	152.53 km ²
Literacy Rate (%)	87.39
Per Capita Income (Rs.) at 2004-05 constant price	38968
Unemployment Rate, 2011-12	0.00
% of households with access to tap water	69.81
% of households with access to electricity	95.84
% of households having toilet facilities within premises	81.15
% of household Waste water outlet connected to drainage	91.11
Type of sewerage system	Underground sewerage system
Type of solid Waste system	Door to Door
% of households with access to computer/laptop with internet	8.23

% of households with access to computer/laptop without internet	12.46
% of households with access to mobile phones	57.36
No. of Hospitals per 1,00,000 people	0.28
No of Schools per 1,00,000 people	54
No of Colleges per 1,00,000 people	4
Swatch Bharat ranking (2015)	25

8. Visakhapatnam, Andhra Pradesh

Indicator	City (Municipal Corporation)
Total Population	1728128
Area	513.61 sq.km
Literacy Rate (%)	81.79
Per Capita Income (Rs.) at 2004-05 constant price	50580
Unemployment Rate, 2011-12	3.95
% of households with access to tap water	71.18
% of households with access to electricity	96.86
% of households having toilet facilities within premises	83.58
% of household Waste water outlet connected to drainage	92.29
Type of sewerage system	Underground sewerage system
Type of solid Waste system	Door to Door
% of households with access to computer/laptop with internet	8.93
% of households with access to computer/laptop without internet	11.39
% of households with access to mobile phones	66.90
No. of Hospitals per 1,00,000 people	3

No of Schools per 1,00,000 people	27
No of Colleges per 1,00,000 people	2
Swatch Bharat ranking (2015)	205

9. Solapur, Maharashtra

Indicator	City (Municipal Corporation)
Total Population	951,558
Area	178.57
Literacy Rate (%)	82.80
Per Capita Income (Rs.) at 2004-05 constant price	45,859
Unemployment Rate, 2011-12	1.65
% of households with access to tap water	96.92
% of households with access to electricity	93.64
% of households having toilet facilities within premises	58.67
% of household Waste water outlet connected to drainage	91.46
Type of sewerage system	Underground sewerage system
Type of solid Waste system	Door to Door
% of households with access to computer/laptop with internet	4.62
% of households with access to computer/laptop without internet	9.08
% of households with access to mobile phones	61.04
No. of Hospitals per 1,00,000 people	2
No of Schools per 1,00,000 people	74
No of Colleges per 1,00,000 people	15
Swatch Bharat ranking (2015)	54

10. Davangere, Karnataka

Indicator	City (Municipal Corporation)
Total Population	434971
Area	77.12 Sq. Km
Literacy Rate (%)	84.90
Per Capita Income (Rs.) at 2004-05 constant price	30219
Unemployment Rate, 2011-12	3.64
% of households with access to tap water	82.45
% of households with access to electricity	97.10
% of households having toilet facilities within premises	82.34
% of household Waste water outlet connected to drainage	94.85
Type of sewerage system	Underground sewerage system
Type of solid Waste system	Door to Door
% of households with access to computer/laptop with internet	5.10
% of households with access to computer/laptop without internet	12.95
% of households with access to mobile phones	65.01
No. of Hospitals per 1,00,000 people	23
No of Schools per 1,00,000 people	133
No of Colleges per 1,00,000 people	16
Swatch Bharat ranking (2015)	NA

11. Indore, Madhya Pradesh

Indicator	City (Municipal Corporation)
Total Population	1964086
Total Population of UA	2170295

Area	172.39 Sq. Km
Literacy Rate (%)	85.87
Per Capita Income (Rs.) at 2004-05 constant price	52501
Unemployment Rate, 2011-12	2.07
% of households with access to tap water	73.74
% of households with access to electricity	98.48
% of households having toilet facilities within premises	90.14
% of household Waste water outlet connected to drainage	92.14
Type of sewerage system	Underground sewerage system
Type of solid Waste system	Door to Door
% of households with access to computer/laptop with internet	11.21
% of households with access to computer/laptop without internet	12.61
% of households with access to mobile phones	62.98
No. of Hospitals per 1,00,000 people	1
No of Schools per 1,00,000 people	21
No of Colleges per 1,00,000 people	5
Swatch Bharat ranking (2015)	N/A

12. New Delhi, NDMC (Delhi)

Indicator	City (Municipal Corporation)
Total Population	257803
Total Population of UA	16349831
Area	42.74 Sq. Km.
Literacy Rate (%)	89.83
Per Capita Income (Rs.) at 2004-05 constant price	112510

Unemployment Rate, 2011-12	2.03
% of households with access to tap water	98.10
% of households with access to electricity	98.82
% of households having toilet facilities within premises	85.03
% of household Waste water outlet connected to drainage	97.06
Type of sewerage system	Underground sewerage system
Type of solid Waste system	Door to Door
% of households with access to computer/laptop with internet	27.74
% of households with access to computer/laptop without internet	16.48
% of households with access to mobile phones	62.65
Swatch Bharat ranking (2015)	16

13. Coimbatore, Tamil Nadu

Indicator	City (Municipal Corporation)
Total Population	1050721
Total Population of UA	2136916
Area	105.6 Sq. Km.
Literacy Rate (%)	91.30
Per Capita Income (Rs.) at 2004-05 constant price	65781
Unemployment Rate, 2011-12	3.21
% of households with access to tap water	98.21
% of households with access to electricity	98.32
% of households having toilet facilities within premises	82.50
% of household Waste water outlet connected to drainage	89.58

Type of sewerage system	Underground sewerage system
Type of solid Waste system	Door to Door
% of households with access to computer/laptop with internet	11.69
% of households with access to computer/laptop without internet	11.56
% of households with access to mobile phones	69.45
No. of Hospitals per 1,00,000 people	0.48
No of Schools per 1,00,000 people	25
No of Colleges per 1,00,000 people	4
Swatch Bharat ranking (2015)	196

14. Kakinada, Andhra Pradesh

Indicator	City (Municipal Corporation)
Total Population	312538
Total Population of UA	443028
Area	57.36 Sq. Km.
Literacy Rate (%)	80.62
Per Capita Income (Rs.) at 2004-05 constant price	37712
Unemployment Rate, 2011-12	4.94
% of households with access to tap water	83.29
% of households with access to electricity	97.40
% of households having toilet facilities within premises	82.63
% of household Waste water outlet connected to drainage	84.14
Type of sewerage system	Underground sewerage system
Type of solid Waste system	Door to Door
% of households with access to computer/laptop with internet	6.08

% of households with access to computer/laptop without internet	9.39
% of households with access to mobile phones	57.46
No. of Hospitals per 1,00,000 people	0.3
No of Schools per 1,00,000 people	134
No of Colleges per 1,00,000 people	9
Swatch Bharat ranking (2015)	NA

15. Belgavi, Karnataka

Indicator	City (Municipal Corporation)
Total Population	488157
Total Population of UA	610350
Area	99.61 Sq. Km.
Literacy Rate (%)	89.82
Per Capita Income (Rs.) at 2004-05 constant price	28856
Unemployment Rate, 2011-12	6.50
% of households with access to tap water	90.14
% of households with access to electricity	97.59
% of households having toilet facilities within premises	91.44
% of household Waste water outlet connected to drainage	88.82
Type of sewerage system	Underground sewerage system
Type of solid Waste system	Door to Door
% of households with access to computer/laptop with internet	8.58
% of households with access to computer/laptop without internet	12.96
% of households with access to mobile phones	61.57

No. of Hospitals per 1,00,000 people	20
No of Schools per 1,00,000 people	97
No of Colleges per 1,00,000 people	13
Swatch Bharat ranking (2015)	NA

16. Udaipur, Rajasthan

Indicator	City (Municipal Corporation)
Total Population	451100
Total Population of UA	474531
Area	56.92 Sq. Km.
Literacy Rate (%)	89.66
Per Capita Income (Rs.) at 2004-05 constant price	24135
Unemployment Rate, 2011-12	0.30
% of households with access to tap water	80.91
% of households with access to electricity	98.59
% of households having toilet facilities within premises	93.26
% of household Waste water outlet connected to drainage	97.85
Type of sewerage system	Underground sewerage system
Type of solid Waste system	Door to Door
% of households with access to computer/laptop with internet	12.48
% of households with access to computer/laptop without internet	14.60
% of households with access to mobile phones	61.13
No. of Hospitals per 1,00,000 people	1
No of Schools per 1,00,000 people	190
No of Colleges per 1,00,000 people	13

Swatch Bharat ranking (2015)	417
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17. Guwahati, Assam

Indicator	City (Municipal Corporation)
Total Population	957352
Total Population of UA	962334
Area	219.06 Sq. Km.
Literacy Rate (%)	91.47
Per Capita Income (Rs.) at 2004-05 constant price	43278
Unemployment Rate, 2011-12	4.32
% of households with access to tap water	32.64
% of households with access to electricity	92.94
% of households having toilet facilities within premises	85.04
% of household Waste water outlet connected to drainage	76.93
Type of sewerage system	Underground sewerage system
Type of solid Waste system	Door to Door
% of households with access to computer/laptop with internet	11.96
% of households with access to computer/laptop without internet	21.82
% of households with access to mobile phones	78.88
No. of Hospitals per 1,00,000 people	1
No of Schools per 1,00,000 people	53
No of Colleges per 1,00,000 people	14
Swatch Bharat ranking (2015)	51

18. Chennai, Tamil Nadu

Indicator	City (Municipal Corporation)
Total Population	4646732

Total Population of UA	8653521
Area	175 Sq. Km
Literacy Rate (%)	90.18
Per Capita Income (Rs.) at 2004-05 constant price	57706
Unemployment Rate, 2011-12	2.32
% of households with access to tap water	84.08
% of households with access to electricity	99.08
% of households having toilet facilities within premises	94.57
% of household Waste water outlet connected to drainage	97.27
Type of sewerage system	Underground sewerage system
Type of solid Waste system	Door to Door
% of households with access to computer/laptop with internet	19.64
% of households with access to computer/laptop without internet	12.51
% of households with access to mobile phones	62.25
No. of Hospitals per 1,00,000 people	1
No of Schools per 1,00,000 people	71
No of Colleges per 1,00,000 people	2
Swatch Bharat ranking (2015)	61

19. Ludhiana, Punjab

Indicator	City (Municipal Corporation)
Total Population	1618879
Area	159.37 Sq. Km.
Literacy Rate (%)	85.77
Per Capita Income (Rs.) at 2004-05 constant price	51633

Unemployment Rate, 2011-12	0
% of households with access to tap water	80.93
% of households with access to electricity	98.89
% of households having toilet facilities within premises	92.58
% of household Waste water outlet connected to drainage	89.47
Type of sewerage system	Underground sewerage system
Type of solid Waste system	Door to Door
% of households with access to computer/laptop with internet	11.75
% of households with access to computer/laptop without internet	10.55
% of households with access to mobile phones	65.30
No. of Hospitals per 1,00,000 people	0.06
No of Schools per 1,00,000 people	28
No of Colleges per 1,00,000 people	5
Swatch Bharat ranking (2015)	381

20. Bhopal, Madhya Pradesh

Indicator	City (Municipal Corporation)
Total Population	1,798,218
Total Population of UA	1,886,100
Area	285.88 Sq. Km.
Literacy Rate (%)	83.47
Per Capita Income (Rs.) at 2004-05 constant price	47214
Unemployment Rate, 2011-12	1.53
% of households with access to tap water	78.58
% of households with access to electricity	97.12

% of households having toilet facilities within premises	82.12
% of household Waste water outlet connected to drainage	91.92
Type of sewerage system	Underground sewerage system
Type of solid Waste system	Door to Door
% of households with access to computer/laptop with internet	10.16
% of households with access to computer/laptop without internet	12.63
% of households with access to mobile phones	61.24
No. of Hospitals per 1,00,000 people	1
No of Schools per 1,00,000 people	66
No of Colleges per 1,00,000 people	12
Swatch Bharat ranking (2015)	106